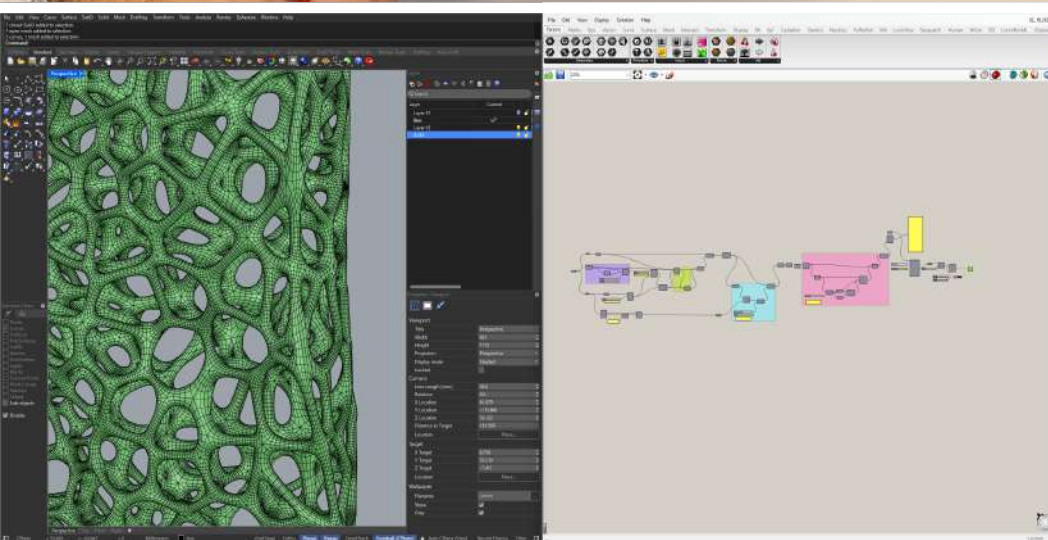
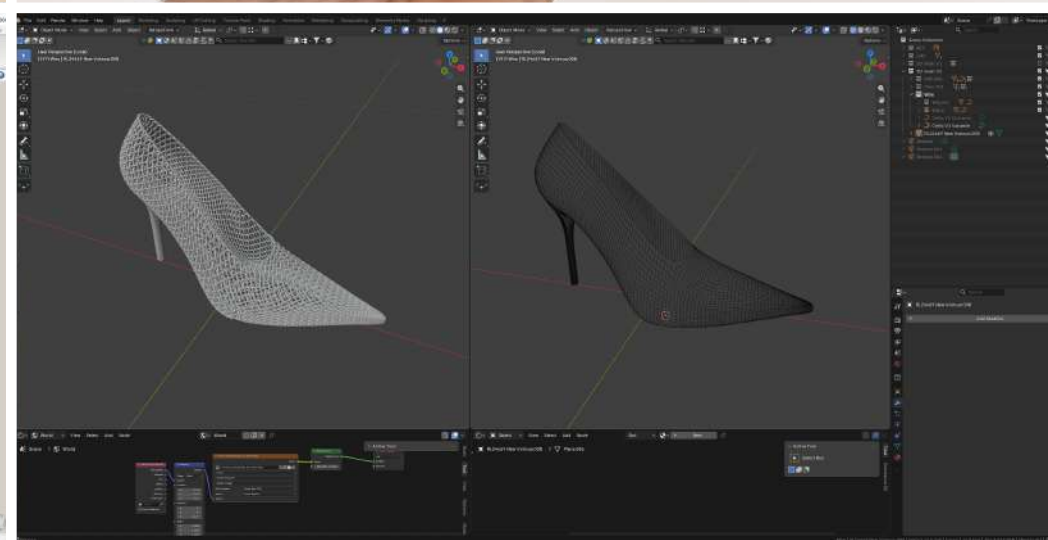


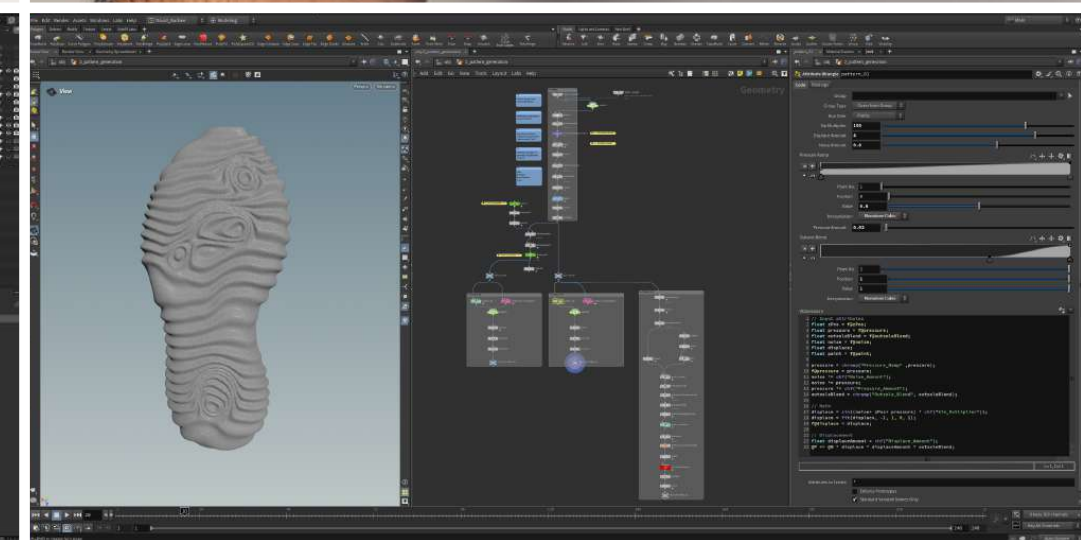
RHINO • GRASSHOPPER



BLENDER



HOUDINI

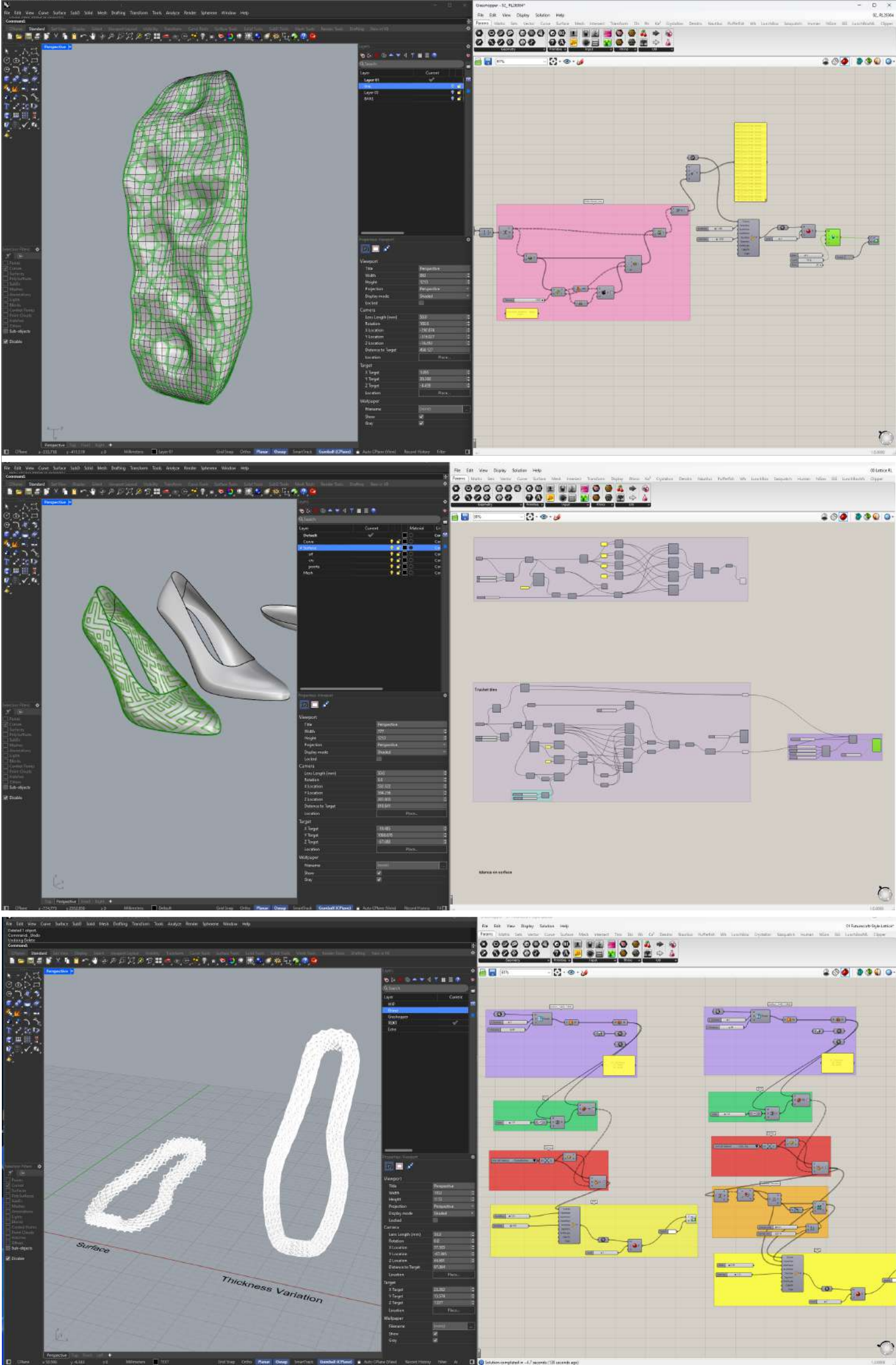


Rhino & Grasshopper

I have structured a suite of parametric tools to automate the creation of complex infills (Lattice and Voronoi), 3D patterns, and textures. The core logic relies on input flexibility, with algorithms specifically categorized to operate on:

- **Curves** (from graphic design)
- **Surfaces** (from CAD design)
- **Meshes** (from polygonal modeling)

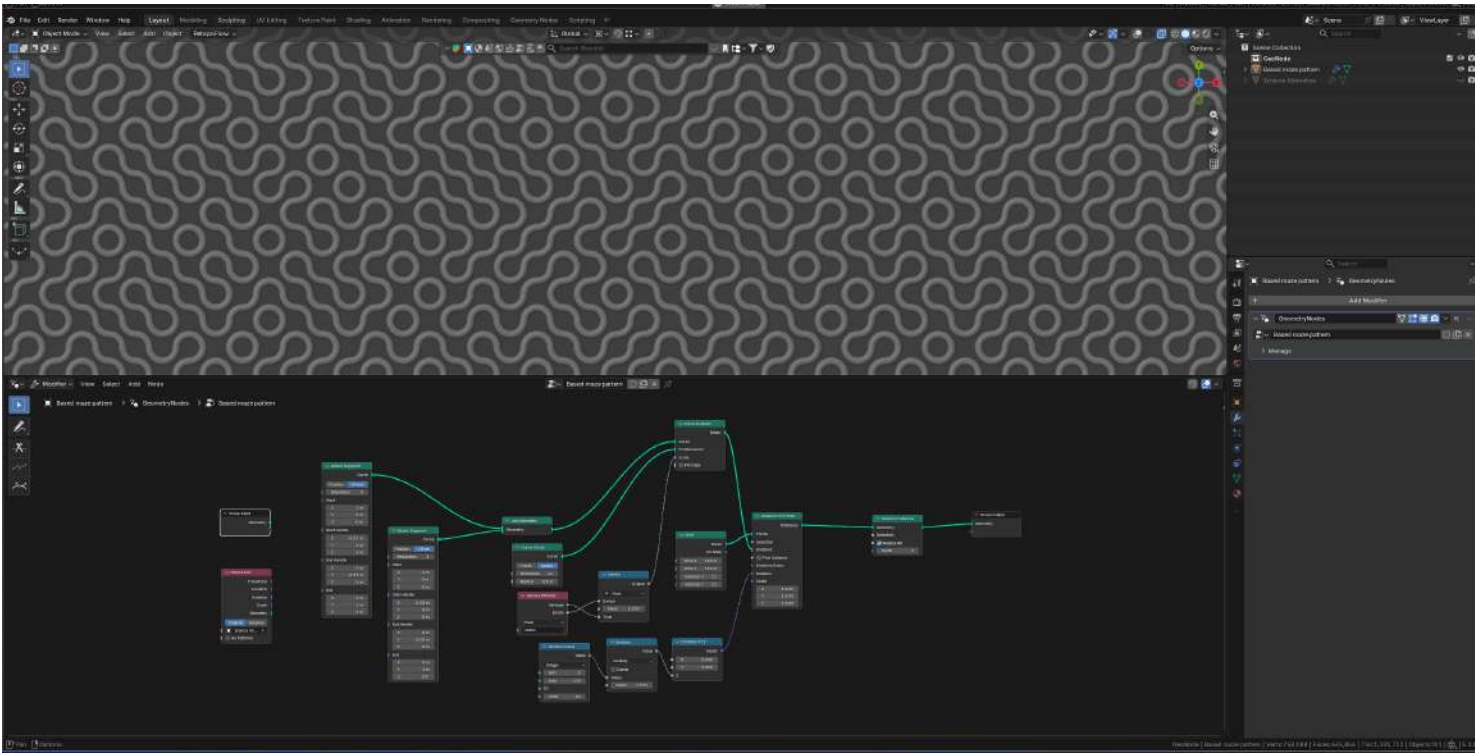
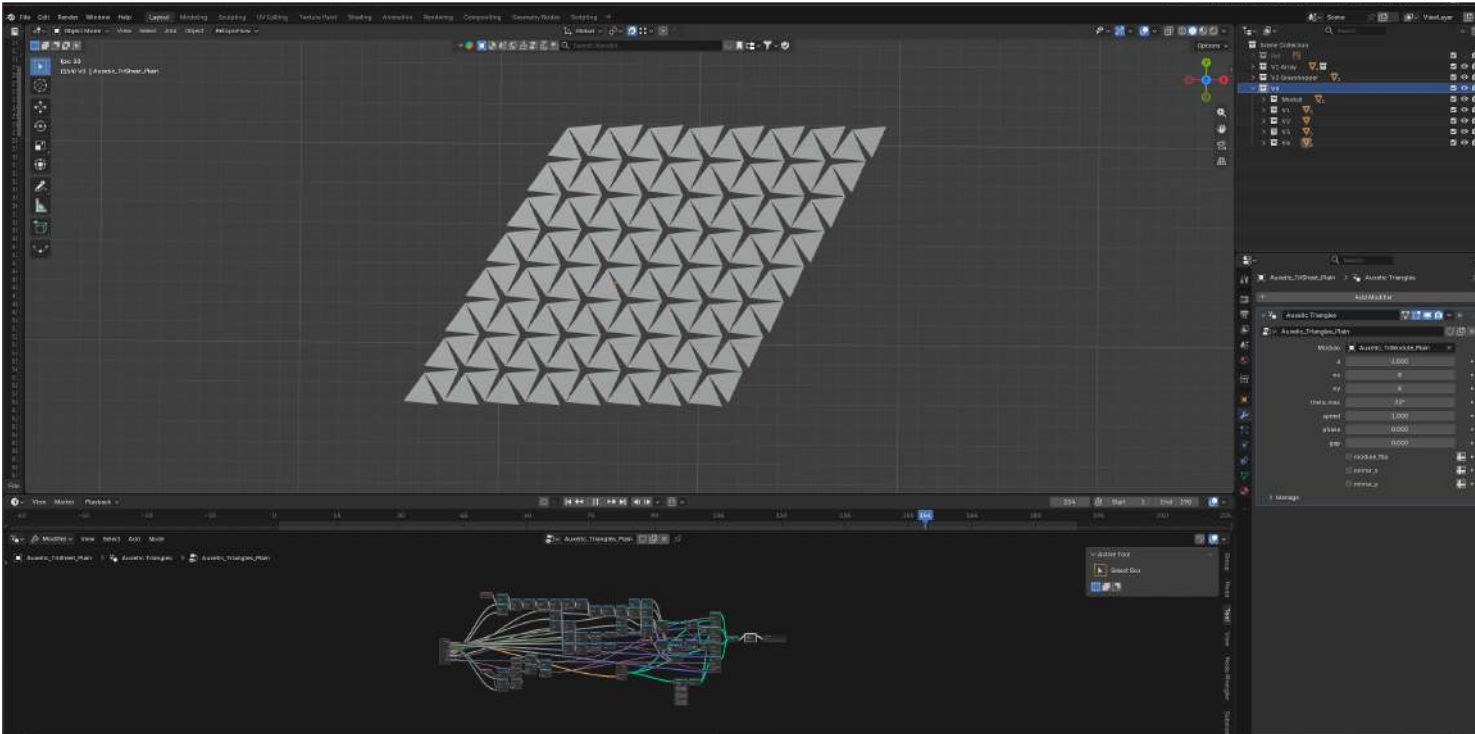
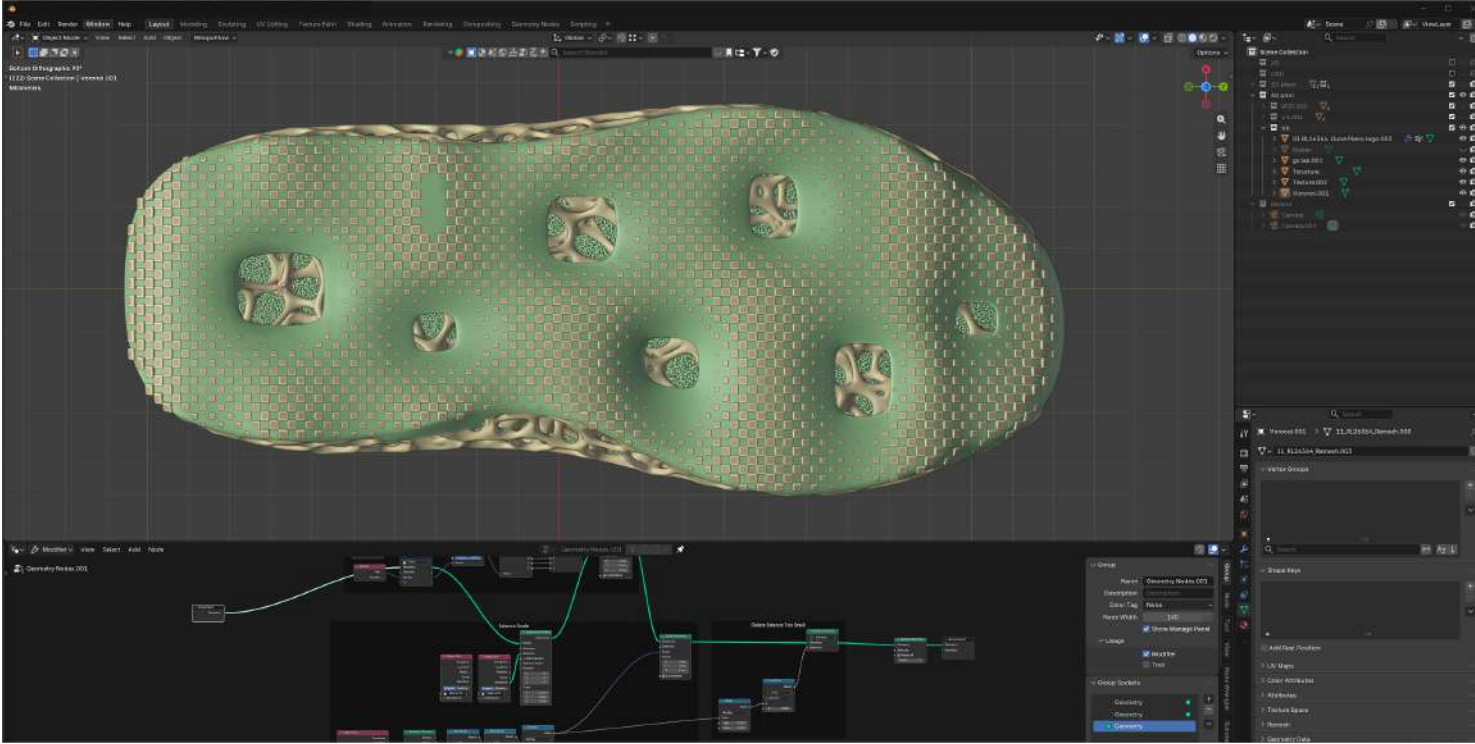
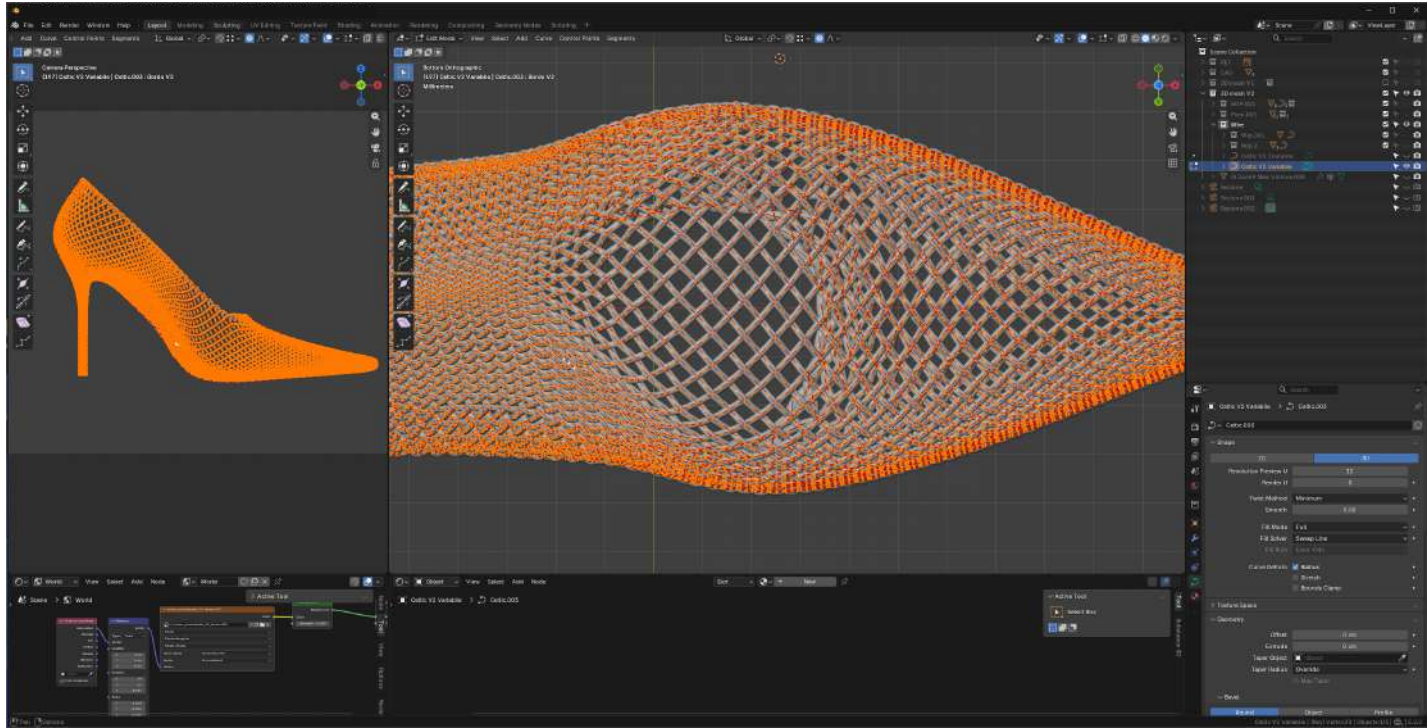
This modular workflow seamlessly adapts to the nature of the source file, creating a direct link between diverse design disciplines and the final printed product.



Blender

As my main software for Sub-D modeling, I use Blender to combine organic design with procedural, node-based logic. My workflow makes extensive use of Geometry Nodes with a focus on:

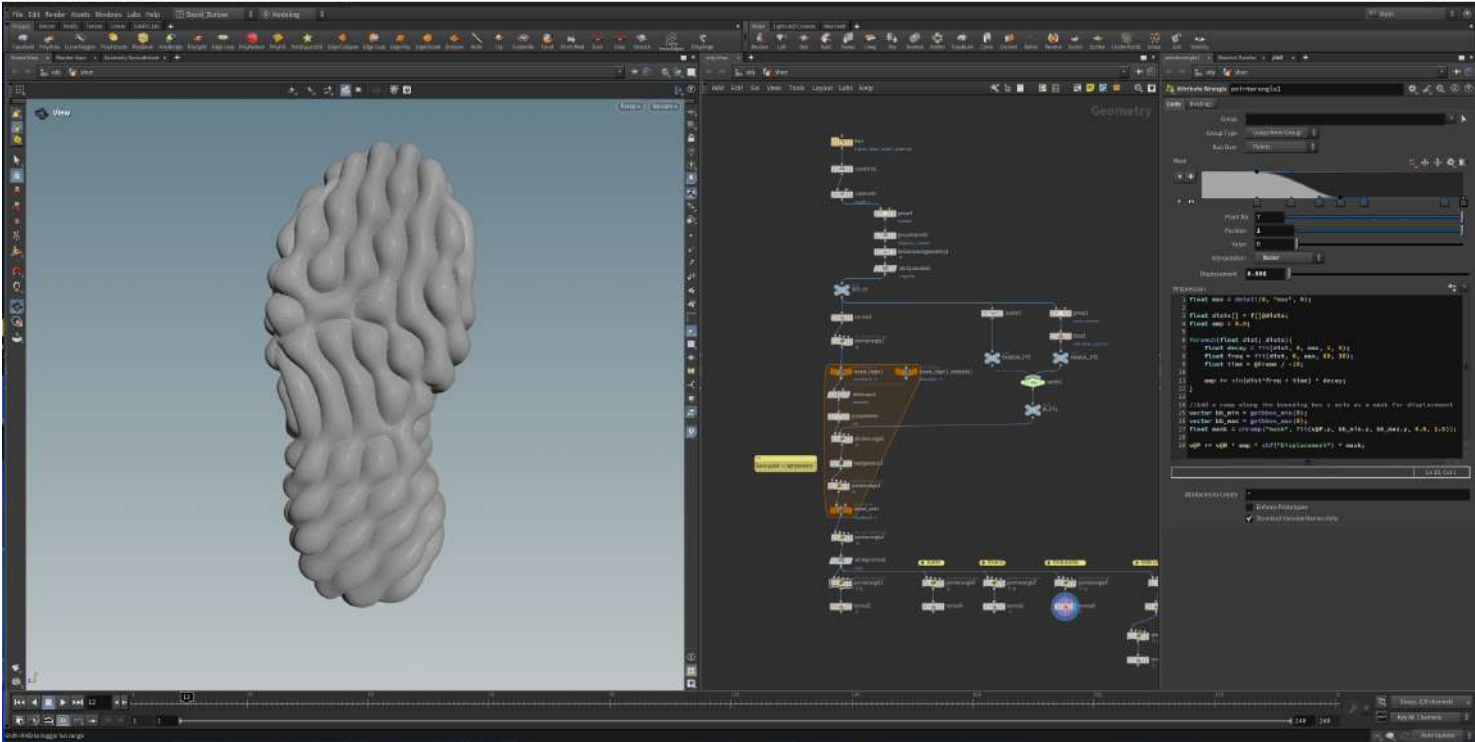
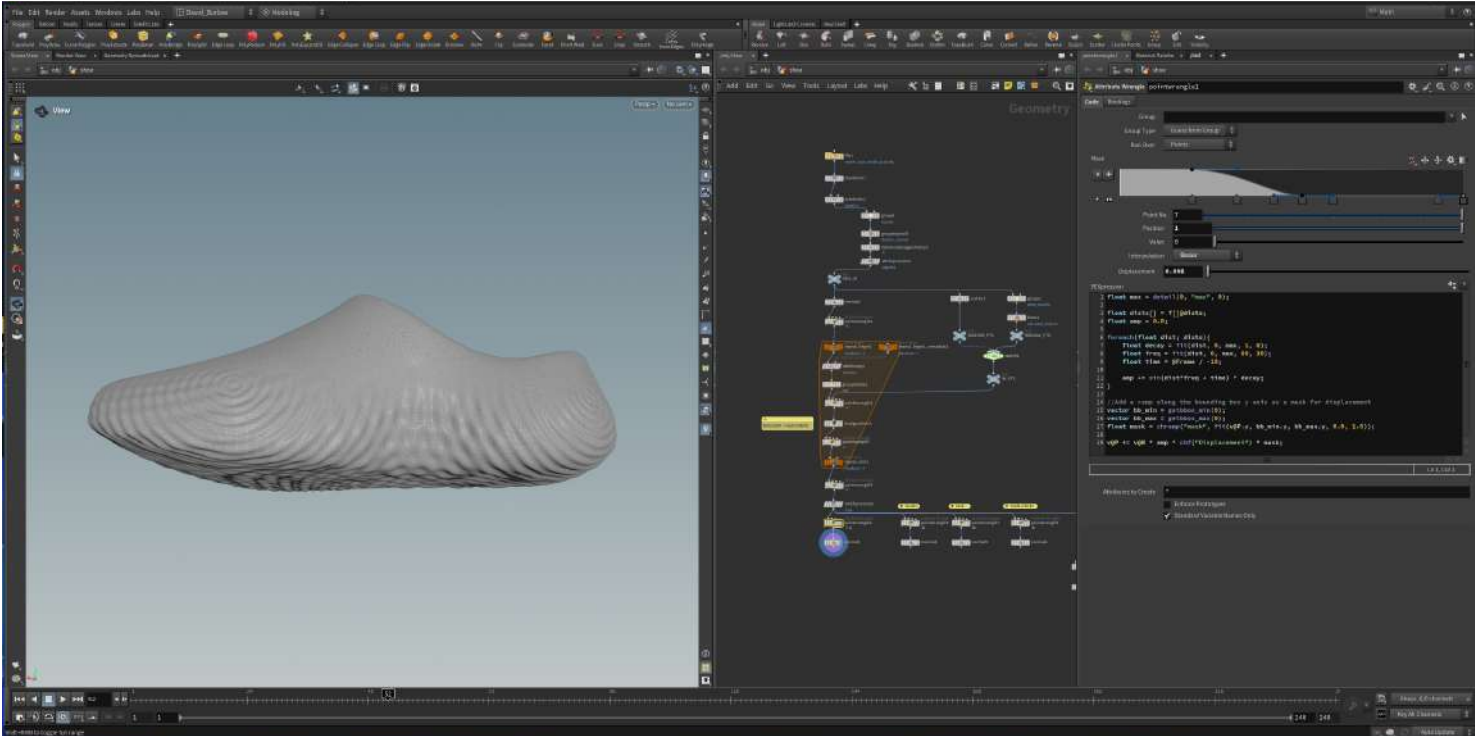
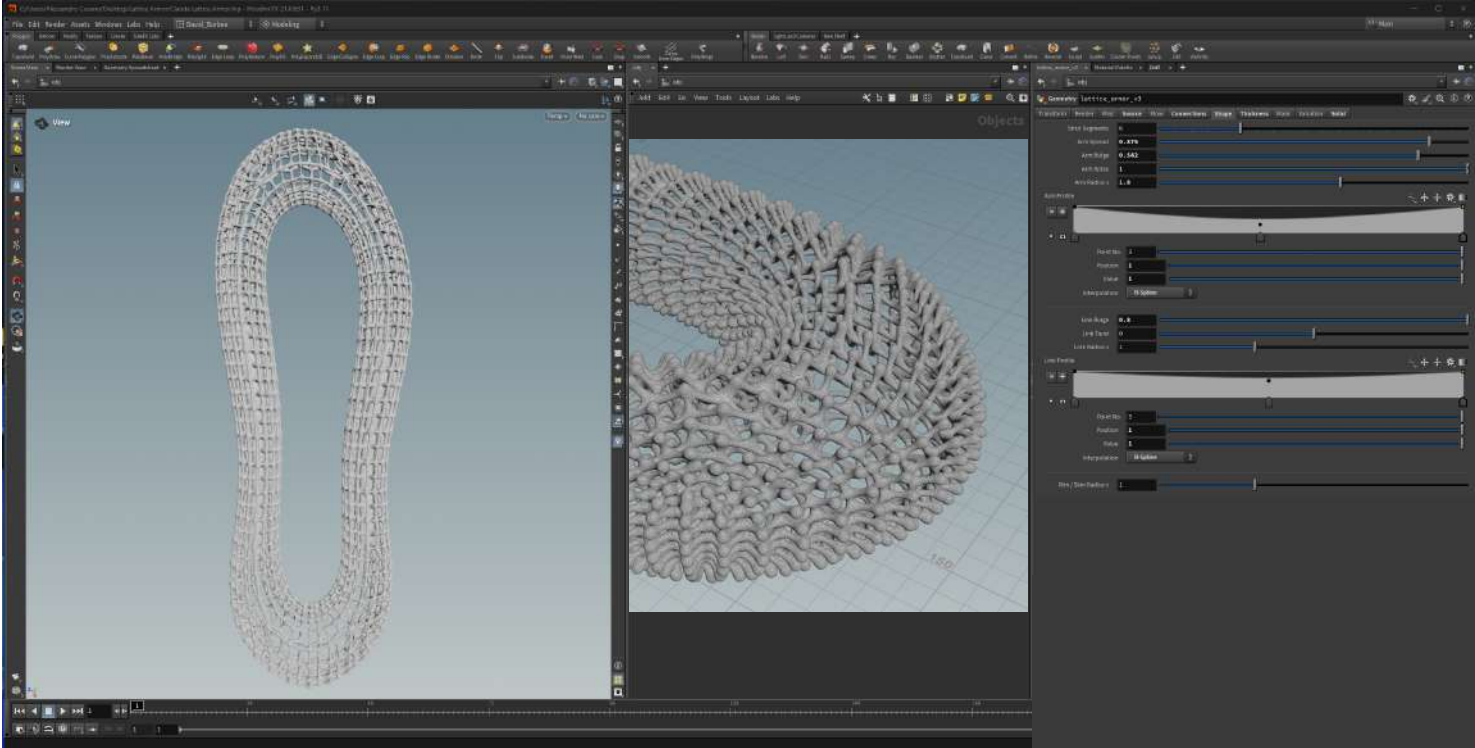
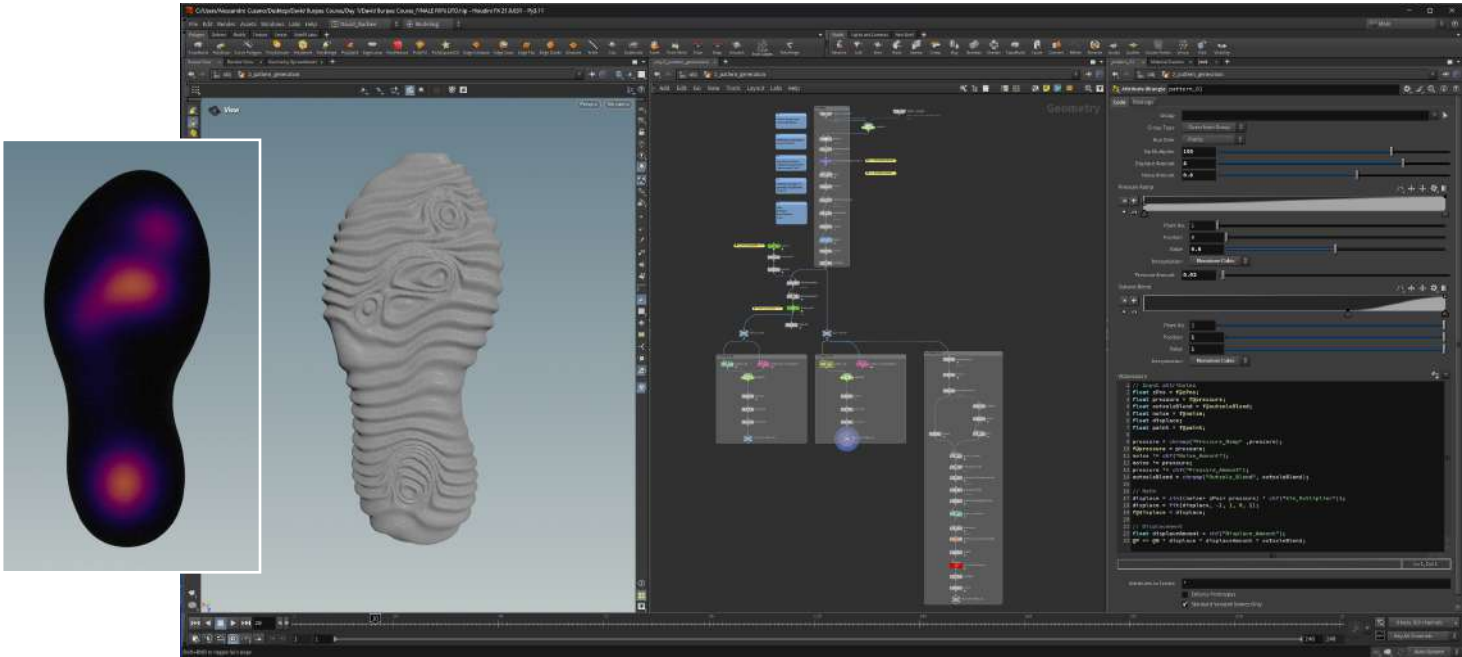
- **Proximity-based instancing:** dynamic control of the distribution, density, and scaling of objects through spatial relationships.
- **Animated texturing:** generating complex, time-driven material behaviors, such as Auxetic structures and Reactive diffusion.
- **Curve modeling:** manipulating complex curve configurations with variable and highly controlled radius to generate complex structures



Houdini

The shift from polygonal workflows to VDB (volumetric) environments ensures seamless integration with the simulation and remeshing of complex geometries. My current research and development work focuses on:

- **Data-driven pattern:** generation of variable topologies mapped to specific data inputs, such as foot pressure points.
- **Procedural Simulations:** create wave interference patterns and use these interactions to generate complex textures.
- **Custom G-Code:** to bypass standard slicers and push commercial FDM printers (Bambu Lab) to their extreme hardware limits.







Thank you for the attention